

Analysis and Design of an Audio Continuous-Time 1-X FIR-MASH Delta–Sigma Modulator

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Abstract—We combine a first-order single-bit CT $\Delta\Sigma$ employing finite impulse-response (FIR) feedback with a 1-bit second-order $\Delta\Sigma$ back end to achieve a modulator with maximum stable amplitude (MSA) that is close to full scale, and a third-order overall noise transfer function (NTF). FIR feedback is used in the input stage to reduce clock-jitter sensitivity, improve linearity, and reduce chopping artifacts. We show that in a MASH ADC, FIR feedback has the additional benefit of filtering the error waveform of the first stage that is fed into the second stage. We apply the principle to an audio continuous-time delta–sigma modulator. A prototype chip, fabricated in 180-nm CMOS to demonstrate the principle, achieves 100.9-dB SNDR in a 24-kHz bandwidth and dissipates 265 μ W. The resulting Schreier figure of merit is 180.5 dB.

Index Terms—Finite impulse-response (FIR) filter, high-resolution, MASH, noise canceling, noise-shaping, oversampling.

I. INTRODUCTION

SEVERAL high-resolution single-loop CT $\Delta\Sigma$ s targeting audio bandwidths have been reported in recent years. Of these, the designs that achieve high power efficiency, as quantified by the Schreier figure of merit (FoM), have used single-bit quantizers in the loop. The performance of a conventional 1-bit CT $\Delta\Sigma$ will be degraded by clock jitter, as well as the nonlinearity of the input integrator due to the rail-to-rail feedback waveform. Fortunately, the use of finite impulse-response (FIR) feedback [1]–[3] makes it possible to address these problems. The idea behind this approach is the following. The single-bit output of the ADC is filtered by an FIR filter with a transfer function $F(z)$, before exciting the main feedback DAC. The output of the FIR DAC is a waveform with many levels, such as in a CT $\Delta\Sigma$ with multibit feedback. Consequently, the modulator's jitter tolerance and linearity are improved. It turns out that FIR feedback also facilitates the use of chopping in the input stage of the CT $\Delta\Sigma$, thereby mitigating flicker-noise problems [4].

While FIR feedback addresses most issues associated with 1-bit operation in a high-order single-loop CT $\Delta\Sigma$, the use of the 2-level quantizer limits the maximum-stable amplitude (MSA) of the modulator. For example, a third-order single-bit CT $\Delta\Sigma$ achieves an MSA of about -3.3 dBFS

(assuming an NTF with an out-of-band gain of 1.5). A small MSA results in increased power dissipation due to the following. The peak SNR of the CT $\Delta\Sigma$ is limited by thermal noise. A smaller MSA would need a lower input-referred thermal noise floor to achieve the desired SNR, resulting in lower impedance levels in the input stage of the CT $\Delta\Sigma$. Consequently, the current draw increases. For instance, an MSA reduction of 10% would need an impedance level that is about 20% lower to restore the peak SNR, thereby increasing power dissipation by 20%. We thus see that if one could somehow increase the MSA of a single-bit CT $\Delta\Sigma$ from -3 dBFS to ≈ 0 dBFS, the Schreier FoM could be improved by about 3 dB.

It turns out that a first-order 1-bit $\Delta\Sigma$ modulator with a noise transfer function (NTF) $\text{NTF}(z) = (1 - z^{-1})$ has an MSA that is close to full scale. This makes intuitive sense due to the following. The transfer function from the quantization noise to the quantizer input, which is given by $\text{NTF}(z) - 1$, is simply z^{-1} , whose magnitude is unity. This is in contrast to that in a high-order $\Delta\Sigma$ loop, where the gain is much larger. Simulations indicate that a first-order 1-bit CT $\Delta\Sigma$ has an MSA of about -0.2 dBFS while achieving a peak SQNR of about 64 dB (with $\text{OSR} = 128$). On the other hand, a third-order modulator whose NTF is chosen to have $\text{OBG} = 1.5$ (in accordance with Lee's rule) has an MSA of -3.3 dBFS but a peak SQNR of 110 dB.

The discussion above suggests that one could obtain a $\Delta\Sigma$ modulator with a high MSA and high SQNR by using a first-order front end (which yields the desired high MSA but results in a high in-band quantization noise) and canceling its quantization noise using a back-end ADC. This is the motivation behind the 1-X MASH CT $\Delta\Sigma$ attempted in this work, where we use a cascade of a first-order 1-bit input stage, followed by a second-order single-bit CT $\Delta\Sigma$ as the back-end ADC. Like other CT $\Delta\Sigma$ s employing a 1-bit quantizer, the 1-X MASH structure also benefits from FIR feedback in terms of jitter sensitivity, loop-filter linearity, and chopping-artifact reduction. Moreover, as we demonstrate in this work, the FIR feedback has an additional advantage in a MASH architecture; it can be used to filter the error waveform that needs to be processed in the second stage of the cascade. This way, the error waveform generated by the first stage not only has much lesser high-frequency content but also has a smaller peak-to-peak value. One can, therefore, use a higher scaling factor between the stages than would otherwise be possible. This MASH structure, which we term the 1-X FIR-MASH converter, is the subject of this article, which is an extended

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version of [5]. The rest of this article is organized as follows. In Section II, we discuss the evolution of the 1-X FIR-MASH architecture and derive the transfer functions of the digital noise-canceling filters needed to achieve the targeted high resolution. Several practical nonidealities, such as a RC time-constant variation and finite integrator gain, can potentially degrade the performance of the 1-X FIR-MASH CT $\Delta\Sigma$. Section III discusses these aspects and gives insight into how the analysis can be used to appropriately modify the transfer function of the noise-canceling filter. Circuit design details are given in Section IV. Section V presents measurement results from a prototype 1-X FIR-MASH CT $\Delta\Sigma$ fabricated in a 180-nm CMOS process. The modulator achieves a peak SNDR of 100.9 dB in a 24-kHz bandwidth and dissipates 265 μ W. The resulting Schreier FoM is 180.5 dB. Conclusions are given in Section VI.

II. EVOLUTION OF 1-X FIR-MASH CT $\Delta\Sigma$ S

The input stage of the 1-X MASH modulator is a first-order CT $\Delta\Sigma$, which we term MOD1. The aim of this is to achieve an MSA that is close to full scale. The quantization error introduced by MOD1 is processed by the back-end ADC. The conventional way of extracting the quantization error of MOD1 is shown in Fig. 1(a). v_1 , which is MOD1's output, is converted into an analog waveform $v_1(t)$ using a DAC. The quantizer's input $y(t)$ is subtracted from this waveform to yield $e_1(t)$, which is digitized by the second-stage ADC. There are several problems with this approach when applied to a 1-bit CT $\Delta\Sigma$. The primary issue is the large peak-to-peak value of $e_1(t)$, which can be as high as ± 3 times the full scale, as shown in Fig. 1(b). To avoid saturating the back-end ADC, $e_1(t)$ must be attenuated by a large factor, which increases noise and distortion of the second stage when referred to the input u . Note that the increased dynamic range of $e_1(t)$ is a consequence of continuous-time operation; in a discrete-time $\Delta\Sigma$ converter, the back end would process the samples $e_1[nT_s]$, which lies between -1 and 1 , as indicated by the magenta dots in Fig. 1(b).

The second problem with extracting $e_1(t)$ as shown in Fig. 1(a), is the presence of step discontinuities in $e_1(t)$. These arise when the sequence v_1 is converted into a waveform using the DAC. Step discontinuities are not desirable as they place increased demands on the linearity of the back-end ADC. A third disadvantage is that $y(t)$ swings between $\pm 2 V_{FS}$, where V_{FS} denotes the full scale of the CT $\Delta\Sigma$. Upon scaling the integrator for dynamic range, its unity-gain frequency has to be reduced (and, significantly, so if one intends to restrict the swing further to simplify the design of the OTA used in the integrator). The end result is that noise and distortion from the back-end ADC become larger when referred to u . A final (and perhaps minor) issue is the increased hardware complexity introduced by the extra DAC needed to obtain $e_1(t)$.

A way to avoid the problems discussed earlier is to feed the input forward and sense the output of the integrator, as shown in Fig. 1(c). Recall that $y(t)$ consists of the input u and delayed quantization noise. Due to the input feedforward path, the integrator output ($e_1(t)$) does not contain the input u .

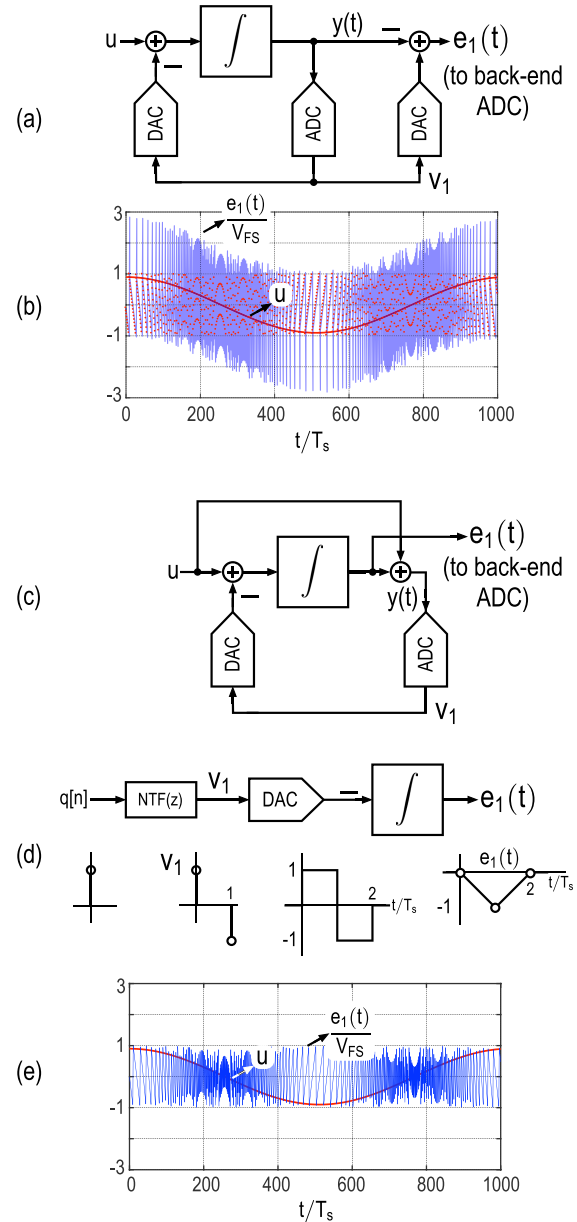


Fig. 1. (a) Extracting the quantization error of MOD1 using an extra DAC. (b) u and $e_1(t)$ normalized to full scale. (c) Error extraction using input feedforward. (d) Signal-processing model. (e) $e_1(t)$ lies between $\pm V_{FS}$ and does not have step discontinuities.

The signal processing model and relevant waveforms depicting the generation of $e_1(t)$ are shown in Fig. 1(d). Assuming that $u = 0$, v_1 is simply the shaped quantization noise of MOD1. The first-order shaped noise is converted into an analog waveform by the DAC and integrated to yield $e_1(t)$. We see that the conversion of the discrete-time sequence $q[n]$ to the continuous-time waveform $e_1(t)$ is characterized by a triangular pulse shape. The samples $e_1[nT_s]$ are simply a one-sample delayed version of the quantization noise sequence $q[n]$. Due to the triangular impulse response from $q[n]$ to $e_1(t)$, the latter is a first-order held waveform whose range is restricted to $\pm V_{FS}$. This, at once, solves two problems with the approach of Fig. 1(a)—not only is the peak-to-peak value of $e_1(t)$ a factor three smaller but also the waveform does not

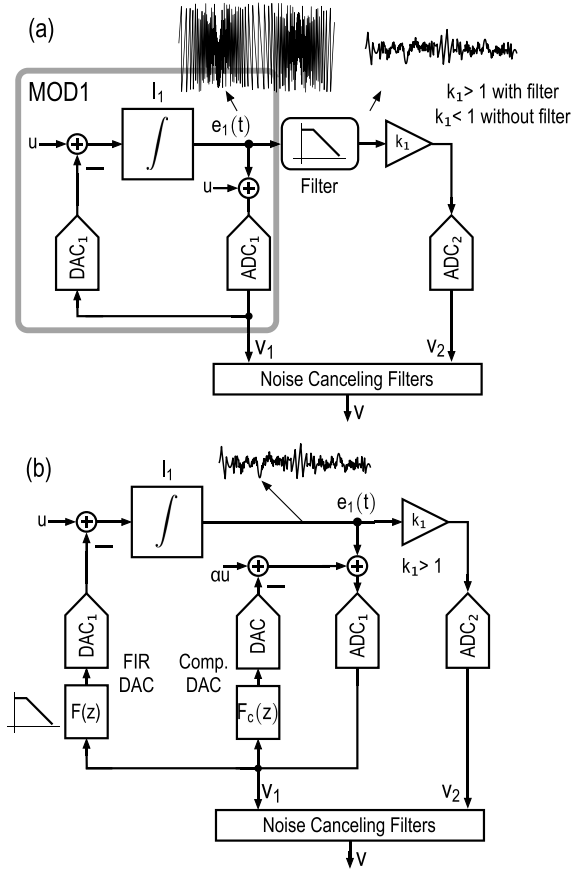


Fig. 2. (a) Filtering MOD1's quantization error, which has significant high-frequency content, to reduce its peak-to-peak value; this enables the use of a higher coupling coefficient and reduces ADC_2 's noise when referred to u . (b) Filter moved into the digital domain through the use of FIR feedback, resulting in the 1-X FIR-MASH CTΔΣM. The FIR DAC reduces clock-jitter sensitivity, improves I_1 's linearity, and enables the use of chopping to eliminate flicker noise.

have step discontinuities. Furthermore, since $e_1(t)$ is also the output of the integrator, the dynamic-range scaling factor can be about three times larger compared with the approach of Fig. 1(a). This is advantageous as it results in a smaller input-referred noise and distortion of the back-end ADC. Finally, unlike in the conventional approach, no extra DAC is needed to generate $e_1(t)$. Using feedforward and tapping off the quantization noise at the integrator output have been proposed as an efficient way of achieving low distortion in discrete-time ΔΣ converters [6], [7]; we see here that the technique has significant additional advantages in the continuous-time context. These positive aspects are further enhanced through the use of FIR feedback, as will be discussed in the following.

Having arrived at an efficient way of extracting the quantization error of MOD1, it is straightforward to conceive of the rest of the MASH converter; the result is shown in Fig. 2(a). For the time being, assume that the filter is bypassed, and $k_1 e_1(t)$ is fed into ADC_2 . In principle, ADC_2 can be implemented with any architecture—in this work, a second-order CTΔΣM is used to realize it. $e_1(t)$ that is the wide-band, full-scale output of the first stage of the cascaded modulator cannot directly excite ADC_2 , as this will saturate the second stage.

The conventional approach to address this problem is to attenuate the error of the first stage [by using $k_1 < 1$, as seen in Fig. 2(a)] before coupling it to ADC_2 . As a result of this attenuation, the quantization noise of the second stage increases when referred to the input of the MASH ADC, thereby degrading its resolution.

An alternative way of reducing the peak signal exciting ADC_2 is to filter the output of I_1 , as shown in Fig. 2(a). Due to the analog low-pass filter, the out-of-band components of $e_1(t)$ are attenuated, resulting in a much smaller, slowly varying signal at its output. The filtered output can then be directly processed by or even be amplified before exciting ADC_2 . Consequently, ADC_2 's quantization noise is reduced by a factor k_1 when referred to the input of the MASH ADC. Introducing the low-pass filter, however, poses a problem. In a CTΔΣM targeting a low signal bandwidth, the area occupied by the filter can become excessive. Besides, the introduction of the filter does not reduce the swing at the output of the integrator. Furthermore, note that the modulator of Fig. 2(a) still has problems with respect to clock jitter and integrator linearity due to the single-bit quantizer employed by MOD1. All these problems are addressed by adding FIR feedback to MOD1, as will be described in the following.

Fig. 2(b) shows the 1-X FIR-MASH architecture, which is an improvement over the design of Fig. 2(a). DAC_1 , which is the main feedback DAC of MOD1, is replaced by an FIR DAC with transfer function $F(z)$. While the structure of Fig. 2(a) filters the quantization error waveform at the output of I_1 , the proposed technique filters the input to the integrator instead. Since this filter is implemented in the digital domain, it is very small and compact even while targeting low signal bandwidths. Furthermore, the use of FIR feedback reduces the sensitivity of the modulator to clock jitter and relaxes I_1 's linearity requirements. The introduction of FIR feedback will modify MOD1's NTF. To restore the NTF to $(1 - z^{-1})$, a compensating path with a transfer function $F_c(z)$ is added around ADC_1 . The input feedforward coefficient α has to be increased to cancel the component of u at the output of I_1 . I_1 's output, which now consists of ADC_1 's quantization error filtered by $F(z)$, has a small peak amplitude. This can, therefore, be directly coupled to ADC_2 . This way the noise of ADC_2 is reduced when referred to the MASH converter's input u .

To summarize, the 1-X FIR-MASH architecture leverages FIR feedback in the input stage of a cascaded CTΔΣM for several benefits. The FIR DAC not only reduces the susceptibility of the MASH converter to clock jitter but also relaxes the linearity requirements of the loop filter. It also enables the use of chopping without causing shaped-noise folding into the signal band. Furthermore, the output of the first stage of the cascade is inherently low-pass filtered, thereby reducing its peak swing. This enables the "residue" to be amplified if necessary, in effect improving the resolution of the second stage.

Fig. 3(a) shows the architecture of a 1-2 FIR-MASH converter, where the back end is a second-order, single-bit CIBF CTΔΣM with an NTF $NTF(z) = (1 - z^{-1})^2$. The sampling rate is assumed to be 1 Hz. ADC_1 and ADC_2 are single bit

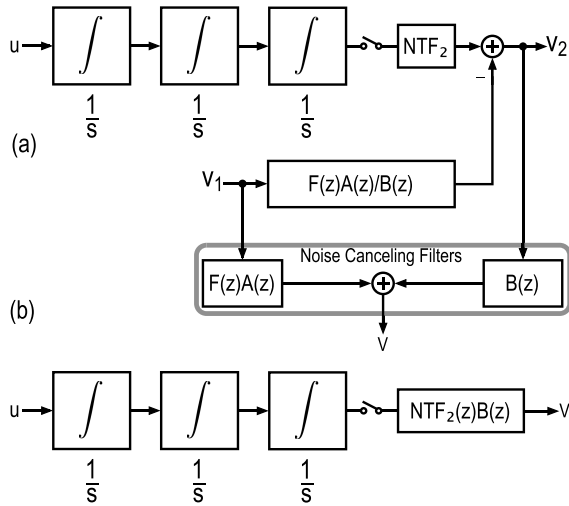


Fig. 4. (a) Signal flow diagram used to determine the signal transfer function of the 1-X FIR-MASH ADC. (b) Equivalent diagram.

seen [from Fig. 4(b)] to be

$$\text{STF}(f) = \frac{1}{(j2\pi f)^3} (1 - e^{-j2\pi f})^3. \quad (6)$$

Though the 1-X FIR-MASH incorporates signal feedforward in the first stage, the overall STF has a third-order roll-off. This makes sense due to the following. The feedforward component adds at the same point as the quantization noise and is canceled after reconstruction. It, thus, does not affect the overall STF. Furthermore, the input passes through three integrators before being sampled in the second stage, explaining the high-order roll-off. Space constraints prevent us from showing the results here but simulations confirm this analysis. The expression (6) is reminiscent of the STF of a single-loop third-order CIFB CT $\Delta\Sigma$ M whose NTF is $(1 - z^{-1})^3$. To summarize, the 1-2 FIR-MASH achieves a third-order NTF given by $(1 - z^{-1})^3$ and an STF that corresponds to that of a single-loop CIFB design. It does this using single-bit quantization and is still able to achieve a maximum stable amplitude (MSA) that is almost full scale.

C. Discussion and Prior Work

Having discussed the motivation behind the 1-X FIR-MASH CT $\Delta\Sigma$ M, we now position this in relation to other CT-MASH topologies described in the literature. Multistage architectures implemented using continuous-time techniques have traditionally been motivated by the quest to achieve medium resolution over very wide bandwidths, which has, in turn, necessitated low OSRs. For instance, the modulator of [9] achieves a peak SNDR of 67 dB over a 465-MHz bandwidth in 28-nm CMOS using a 1-2 CT-MASH architecture. Since the OSR is only 8, the authors are forced to use a 17-level quantizer in MOD1. The quantization error of the first stage is extracted using the conventional technique of Fig. 1(a). Since the quantization error is a small fraction of the full scale (due to multibit operation), the increased peak-to-peak value of the continuous-time quantization error waveform is easily addressed by the second-order CIFB back end.

The design reported in [10] is a 2-1-1 cascaded CT $\Delta\Sigma$ M that achieves a peak SNDR of 62.5 dB in an 18-MHz bandwidth. The authors feed the modulator's input u to the output of the loop filter of the first stage of the cascade. The loop filter's output, therefore, is a signal that does not consist of u . However, it consists of the output of the direct path (needed for excess-delay compensation) and the first- and second-order paths. The addition of the direct and first-order paths increases the high-frequency content and peak-to-peak swing of the sensed error, without contributing to the in-band signal. As a result, successive stages of the cascade are unnecessarily stressed with signals having sharp edges and a higher swing.

Liu *et al.* [11] present a 1-1-1 MASH CT $\Delta\Sigma$ M in 40-nm CMOS that achieves a peak SNDR of 67.3 dB in a 50-MHz bandwidth. It uses a first-order single-bit input stage with FIR feedback. However, the loop is not compensated for the effect of the FIR DAC. The main motivation to use MOD1 appears to be its stable operation even if not exactly compensated. Quantization error of MOD1 is obtained using the traditional technique. In contrast, our work compensates the loop for the effect of the FIR DAC while, at the same time, exploiting FIR feedback to significantly reduce the high-frequency content and peak-to-peak swing of the signal that is processed by the back-end ADC. To our best knowledge, we are not aware of any high-resolution MASH CT $\Delta\Sigma$ Ms.

III. EFFECT OF PRACTICAL NONIDEALITIES

A. RC Variations

Time-constant variation is perhaps the most important non-ideality that degrades the performance of a continuous-time MASH converter, and the first step in addressing this issue is to analyze the effect of RC variations. All integrators in our work are realized using active- RC techniques. RC variation changes the unity-gain frequency of the integrators in the modulator. Though resistors and capacitors are realized using digitally tunable banks and are tuned so as to be well within $\pm 5\%$ of their nominal values, it turns out that this is not adequate to achieve 100-dB SNDR without appropriately modifying the noise-canceling filters.

The mechanism of performance degradation due to RC variation is as follows. A change in the integrator unity-gain frequency causes the transfer function from v_1 to v_2 to vary. Referring to Fig. 3(b) and denoting the normalized unity-gain frequency of the integrators by k_p (nominally 1), we see that $A(z)/B(z)$ in (2) is modified as

$$\frac{\hat{A}(z)}{\hat{B}(z)} = c2d \left[\frac{k_p^3}{s^3} \right] \hat{\text{NTF}}_2(z) \quad (7)$$

where

$$\begin{aligned} \hat{\text{NTF}}_2(z) &= \frac{1}{1 + c2d \left[\frac{1.5k_p}{s} + \frac{k_p^2}{s^2} \right]} \\ &= \frac{(1 - z^{-1})^2}{1 + \underbrace{(0.5k_p^2 + 1.5k_p - 2)z^{-1} + (0.5k_p^2 - 1.5k_p + 1)z^{-2}}_{D(z)}}. \end{aligned} \quad (8)$$

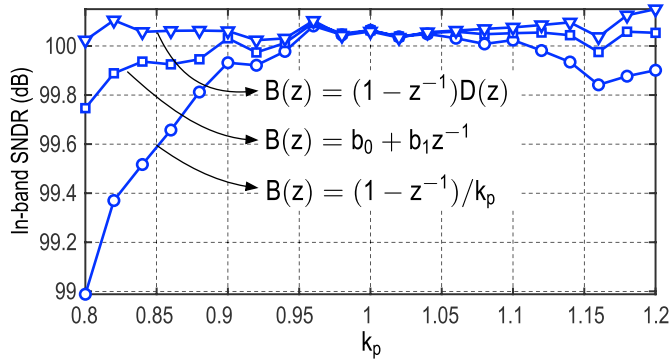


Fig. 5. In-band SNDR of the 1-2 FIR-MASH CT $\Delta\Sigma$ M with different choices of noise-canceling filter transfer functions. In all cases, v_1 is processed by an FIR filter with transfer function $F(z)A(z)$.

Using this in (7), we obtain

$$\frac{\hat{A}(z)}{\hat{B}(z)} = k_p^3 \frac{\frac{1}{6}z^{-1} + \frac{2}{3}z^{-2} + \frac{1}{6}z^{-3}}{(1 - z^{-1})D(z)}. \quad (9)$$

From the discussion above, we see that $\hat{A}(z) = (1/6)z^{-1} + (2/3)z^{-2} + (1/6)z^{-3}$, which is independent of k_p . Thus, the noise-canceling filter that processes v_1 , namely, $F(z)A(z)$, does not change when RC time-constants vary. The filter that processes v_2 , however, needs to be modified according to

$$\hat{B}(z) = \frac{1}{k_p^3}(1 - z^{-1})D(z). \quad (10)$$

Since $D(z)$ has three taps [see (8)] and using (10), it seems as if $\hat{B}(z)$ needs to be a four-tap FIR filter. This, however, is only true if one needs to cancel the quantization noise of MOD1 at all frequencies. If one is content with noise-cancellation only in the signal band, one can expect that a shorter filter should suffice. To confirm this, a 1-2 FIR-MASH CT $\Delta\Sigma$ M with 12-tap FIR feedback was simulated as k_p is varied from 0.8 to 1.2, corresponding to a $\pm 20\%$ change in the unity-gain frequency of the integrators. The input-referred in-band thermal noise of the CT $\Delta\Sigma$ M was chosen as -101 dBFS. A 6-kHz -1 dBFS sinusoid was used as the input. The output streams v_1 and v_2 of the CT $\Delta\Sigma$ M were combined in three different ways. In all three cases, v_1 was filtered by an FIR filter with transfer function $F(z)A(z)$. $B(z)$, however, is chosen in three ways, as described below.

- 1) $B(z) = (1 - z^{-1})D(z)$, where $D(z)$ is chosen according to (8).
- 2) $B(z)$ is chosen as a two-tap FIR filter $b_0 + b_1 z^{-1}$ with b_0 and b_1 chosen (using the LMS algorithm) to minimize in-band noise in v .
- 3) $B(z)$ is chosen as $(1 - z^{-1})/k_p$. This choice arises when one ignores the frequency dependence of $D(z)$ in (10) altogether and replaces $D(z)$ by its dc gain, which is k_p^2 .

Fig. 5 shows the resulting in-band SNDR with each of the abovementioned strategies. As expected, when strategy 1) is used, the SNDR remains virtually the same over a $\pm 20\%$ change in $1/RC$. Using a two-tap $B(z)$ also does almost as nicely and only degrades performance by about 0.2 dB for $k_p = 0.8$. Interestingly, using $B(z) = (1 - z^{-1})/k_p$ results in

only a 1-dB degradation in performance. In practice, a $\pm 20\%$ change in k_p is pessimistic since we employ digitally tunable RC banks to restore the unity-gain frequency of the integrators to well within $\pm 5\%$ of their nominal values. At a particular setting of the RC -bank, the variation in time-constant over temperature is about $\pm 2\%$.

To summarize, if RC time-constant variation was the only nonideality, noise canceling can be achieved by simply choosing $B(z) = (1 - z^{-1})$ and adjusting the gain depending on k_p .

B. Finite Integrator Gain

In practice, the gain of the OTAs used in the integrators is finite, resulting in leaky integration. As a consequence, the NTF of MOD1 does not have zero gain at dc. The dc gain of the back-end ADC is also modified from unity. In general, therefore, the transfer functions of the noise-canceling filters need to be modified to account for finite integrator gain. Space constraints prevent us from giving a detailed analysis here, but it turns out that cancellation can be achieved by continuing to process v_1 with $F(z)A(z)$ and using a two-tap $B(z) = b_0 + b_1 z^{-1}$, as in case 2) of Section III-A. The intuition behind this is the following. Assuming $k_p = 1$ and infinite integrator gain, $B(z) = (1 - z^{-1})$ corresponding to the NTF of MOD1. With leaky integration, $B(z)$ must be modified to $\approx (1 - (1 - 1/A_0)z^{-1})$, where A_0 denotes the integrator's dc gain. When only k_p varies (but with infinite integrator gain), it suffices to simply change the gain of $B(z)$ (see Fig. 5). It, thus, seems reasonable that when both effects are present, we need two independent degrees of freedom in $B(z)$ and reflect as $b_0 + b_1 z^{-1}$.

C. Mismatch and Finite Op-Amp Bandwidth

Mismatch among the elements of the FIR DAC will alter the transfer function $F(z)$ and, thereby, alter the frequency content of the signal being processed by the back end. Due to the high OSR ($=128$) used in this work, the effect of tap mismatch is simply equivalent to a gain error. Resistor and capacitor mismatch in the back-end ADC results in a modified signal transfer function. In the signal band, however, the effect of this is to simply modify the dc gain of the back-end ADC. It, thus, follows that element mismatch, the effect of which is similar to that of a varying k_p , can also be addressed by the two tap FIR noise-canceling filter $B(z) = b_0 + b_1 z^{-1}$.

The OTAs in the active- RC integrators are realized as two-stage feedforward compensated designs. The unity-gain bandwidths of the OTAs are more than ten times the modulator's sampling rate. Finite OTA bandwidth has virtually no effect on the modulator's performance. This is confirmed by simulations.

D. Calibration

There has been interesting prior work on the calibration of continuous-time MASH converters. Background techniques have used the basic idea of injecting a test signal into the input stage of the cascade, correlating it with the reconstructed

output and modifying either the RC -setting of the modulator (see [10] for instance) or the noise-canceling filter in the digital domain [12]. To the best of our knowledge, reported prior calibration techniques have targeted wide-band MASH CT $\Delta\Sigma$ designs achieving moderate resolution. Furthermore, since LMS techniques can require several million samples to converge, it appears as if they are not appropriate for a low-speed high-precision design such as this, since they will most likely take several seconds to converge and may not be able to track temperature changes. The analog-domain correction of RC time constants (as in [10]) is not appropriate in our application that targets high SNDR, as it requires extremely fine-grained tuning of the RC banks. Based on the abovementioned considerations, we adopted the following approach.

- 1) The 1-2 FIR-MASH converter is calibrated under nominal conditions (room temperature) using a two-tap FIR filter for $B(z) = b_0 + b_1z^{-1}$. The intuition behind the need for a two-tap filter was outlined earlier in this section. The weights b_0 and b_1 are determined using the frequency-domain least-squares technique and is detailed in the Appendix. This step accounts for the nominal value of k_p and integrator dc gain, as well as all other factors (such as mismatch) that change the gain of MOD1's quantization noise, as measured by the back-end ADC.
- 2) Temperature variations will cause k_p and integrator dc gain to change and degrade the in-band noise cancellation performance if $B(z)$ is not appropriately modified. The change in k_p is predominantly due to the temperature coefficient of the resistors. A simple, practical way of estimating k_p would be to build an on-chip RC oscillator and compare its frequency with the modulator's sampling rate. Since the primary aim of this article is to test the efficacy of the FIR-MASH architecture, we simply used the resistor temperature coefficient information provided by the foundry to estimate k_p at a given temperature. Integrator dc gain A_0 was estimated by adding a dc input to the quantizer and detecting the change in the average output of MOD1. Other schemes for determining A_0 can be conceived. Since k_p and A_0 are known under a given operating condition, $B(z)$ at a given temperature are modified as follows. We denote $B(z)$ obtained during foreground calibration at room temperature by $b_0 + b_1z^{-1}$, while $k_{p,\text{nom}}$ and $A_{0,\text{nom}}$ denote the nominal values of k_p and A_0 . As temperature varies, $B(z)$ is modified as

$$B(z) = \frac{k_{p,\text{nom}}}{k_p} \left\{ b_0 + \left(b_1 - \frac{b_0}{A_{0,\text{nom}}} + \frac{b_0}{A_0} \right) z^{-1} \right\}. \quad (11)$$

IV. CIRCUIT DESIGN

The simplified single-ended schematic of the 1-2 FIR-MASH CT $\Delta\Sigma$ is shown in Fig. 6(a). Negative resistances refer to inversion in the differential version. The sampling rate is $f_s = 6.144$ MHz. All integrators are realized using active- RC techniques for high linearity and low noise.

The main FIR DAC has 12 identical taps; equal weights not only ease layout but also introduce spectral nulls in the feedback waveform at multiples of $f_s/12$. Because of this, the first stage of A1 can be chopped at $f_s/24$, thereby enabling the modulation of its flicker noise out of the signal band. The FIR DAC is implemented in a semidigital fashion, as shown in the inset of Fig. 6(a). The DAC unit elements are resistive for low noise. Due to the semidigital implementation, DAC unit-element mismatch does not result in nonlinearity; it merely modifies the transfer function of the FIR filter. The effect of this on jitter sensitivity is negligible. The compensating FIR DAC, whose transfer function is $F_c(z)$, is realized in a manner similar to the main DAC and shares the same flip-flops. $F_c(z)$ is derived using the theory in [13]. Ra implements the input feed-in, needed to ensure that the output of A1 does not contain u . As described in Section III, the output of A1 consists of ADC1's quantization noise, filtered by the FIR transfer function $F(z)$. Because of filtering, the unity-gain frequency of MOD1's integrator can be significantly increased compared with a conventional design of the type shown in Fig. 1(a). The increased unity-gain frequency reduces the noise contributions of the rest of MOD1 and the back-end ADC when referred to u . All resistors and capacitors are realized using digitally programmable banks, which can be set so that the RC time-constant realized is well within $\pm 5\%$ of the nominal value. We included a pseudorandom two-level dither generator on the chip, with the provision to inject it at an auxiliary input of the summing amplifier in MOD1 if needed.

The single-bit ADCs are realized using the StrongARM latch-based design shown in Fig. 6(b), where the outputs of the latch excite an RS flip-flop to yield the single-bit decision. Compared with a sense-amplifier-based latch, a StrongARM structure is slower and has a higher dynamic offset. Offset is noise-shaped anyway, and the lower speed is not problematic at the 6.144-MHz sampling rate used in our design. An advantage of the StrongARM latch is the significantly simpler clocking and resulting reduced layout parasitics.

A3 and A4 are the OTAs in the integrators of the back-end converter of the MASH. They form a second-order CIBF CT $\Delta\Sigma$, whose nominal NTF is $(1 - z^{-1})^2$. The impedance levels of the back-end ADC are significantly higher than those in the input stage, resulting in much lower power dissipation in A3 and A4. Rd is implemented using a T-network for the reduced area.

Fig. 7 shows the macromodel and the simplified schematic of the two-stage feedforward-compensated OTA used in the integrators. The input stage G_{m1} is a cascoded differential pair with current reuse. The cascode devices not only enhance the stage's dc gain but also help to reduce the parasitic capacitances at nodes b and b' . G_{m1} is chopped at a frequency $f_c = f_s/24$. G_{m2} , which serves as the high-gain path, and G_{m3} , which forms the feedforward path, are realized using the same differential pair (and current reuse). The simulated dc gain and unity-gain bandwidths (at room temperature) are about 79.6 dB and 64 MHz, respectively. These simulations are done on the postlayout extracted netlist of the op-amp, with a chopping frequency of $f_s/24$. The common-mode feedback (CMFB)

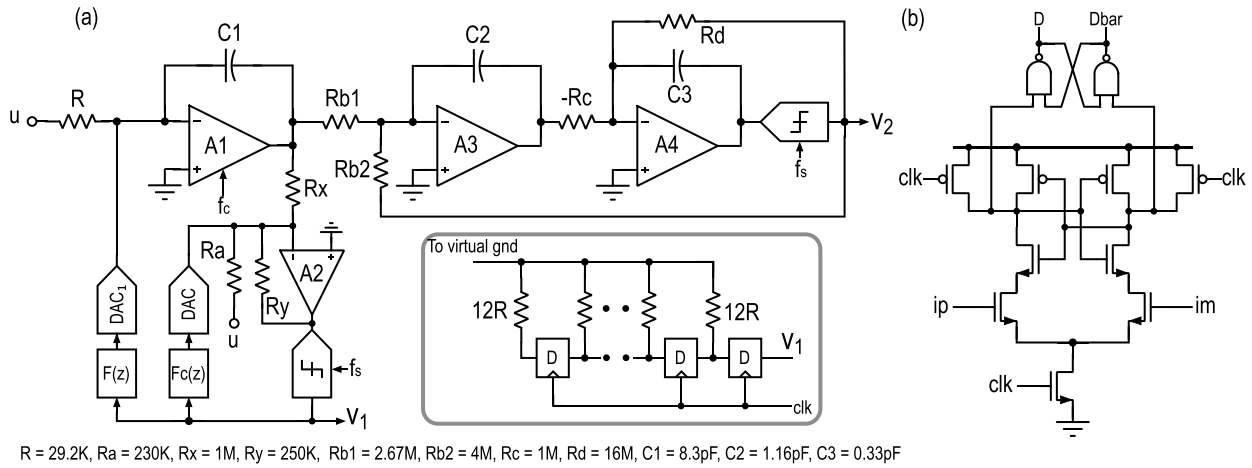


Fig. 6. (a) Simplified single-ended schematic of the 1-2 FIR-MASH CTΔΣM. (b) StrongArm-based single-bit ADC.

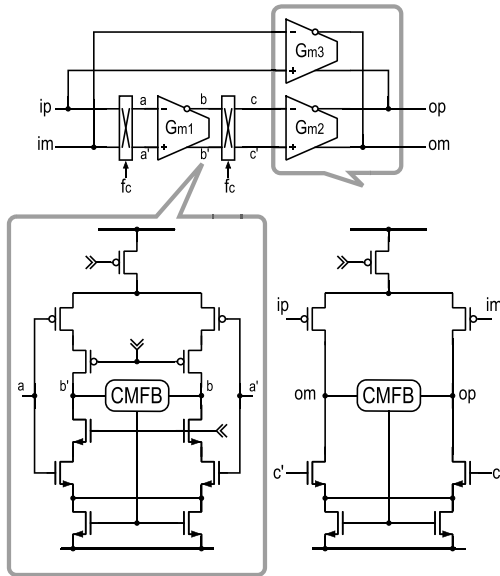


Fig. 7. Simplified schematic of the two-stage feedforward compensated OTA, with the chopped input stage.

circuitry at the output of the first stage uses a differential-pair-based common-mode detector to ensure capacitive loading (so as to not degrade dc gain). Output CMFB uses a resistive detector, which has a large common-mode detection range.

The power breakup of various parts of the chip is as follows. The op-amps and digital portions (including the flip-flops and drivers in the FIR DAC) account for 40% and 38% of the power each, while the remaining 22% is drawn from the references. A1–A4 consume about 65%, 15%, 5%, and 15% of the loop-filter's power, respectively.

A. Flicker Noise Reduction by Chopping

Input-referred flicker noise of the OTA A1 in MOD1 (see Fig. 6) degrades the in-band SNDR of the CTΔΣM. A brute-force way of addressing this is to use large device sizes for the input stage of the OTA. This approach, however, results in a large parasitic capacitance at the OTA's input,

thereby degrading the speed and linearity of the integrator. An alternative technique to solve this problem is to chop the first stage of the OTA. Chopping the OTA at f_c modulates its flicker noise to f_c ; this modulated noise is subsequently eliminated in the decimation filter. Since the flicker noise of the input stage (Gm1 in Fig. 7) is no longer a concern, it can deliberately be made small. This benefits the linearity and speed of the integrator.

The input integrator processes the shaped quantization noise. It turns out, as shown in [4], that chopping causes aliasing of shaped noise from multiples of twice the chopping frequency. The root cause is the abrupt switching of the parasitic input and output capacitances of the OTA's first stage (Gm1 in Fig. 7) at every chopping edge. Because of the FIR feedback, however, the shaped-noise processed by the integrator of MOD1 has spectral nulls at integral multiples of $f_s/12$. Choosing the chopping frequency as $f_c = f_s/24$ will thus downconvert the shaped noise from around the spectral nulls of the feedback waveform, where the power is already very small. As a result, chopping artifacts that can be very problematic with an inappropriate choice of f_c are no longer an issue. During the simulation phase, it becomes necessary to estimate the power of quantization noise that aliases in-band due to chopping. This is done in an efficient way by using the integrator test bench described in [4]. Simulations indicate that power due to chopping artifacts is about 10 dB lower than the in-band quantization noise of the MASH CTΔΣM.

V. MEASUREMENT RESULTS

The 1-2 MASH CTΔΣM was fabricated in a 0.18-μm CMOS process through Europractice. Fig. 8 shows the die photograph and the layout of the active area, which measures about 0.65 mm². For SNDR measurements, the modulator was excited by a 6-kHz sinusoid from an Audio Precision (AP2700) audio analyzer. The output single-bit sequences (v_1 and v_2) of the two stages of the MASH ADC were captured. As shown in Fig. 3(c), v_1 and v_2 have to be filtered by $F(z)A(z)$ and $B(z)$, respectively, and combined. $B(z)$ is computed in foreground fashion off-chip, using least-squares minimization, as described in the Appendix.

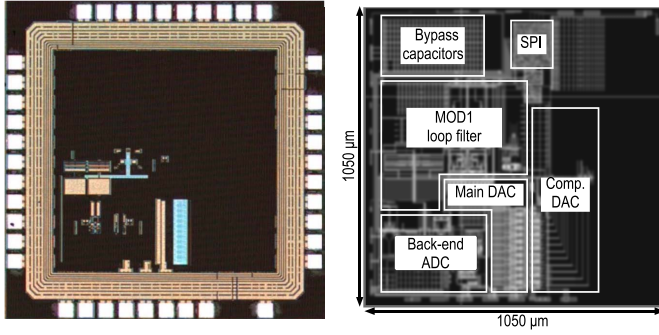


Fig. 8. Chip photograph and layout snapshot.

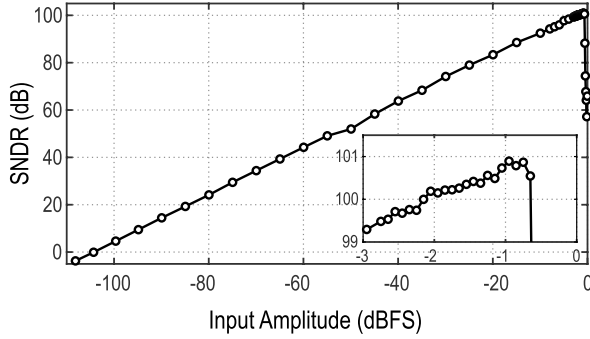


Fig. 9. Measured SNDR as a function of input amplitude.

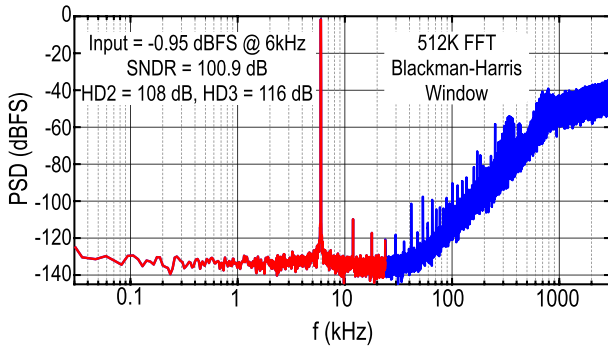
Fig. 10. Measured output PSD for a -0.95 dBFS input.

Fig. 9 shows the measured SNDR as a function of the input amplitude. The peak SNDR is 100.9 dB, and the dynamic range is 104 dB. Due to the use of MOD1 as the first stage, the MSA is as large as -0.65 dBFS, even with a single-bit quantizer. Fig. 10 shows the measured PSD of the MASH CT $\Delta\Sigma$ M when it achieves the peak SNDR. The HD2 and HD3 are 108 and 116 dB, respectively.

Fig. 11 shows the in-band PSD with chopping turned on and off. Recall that the chopping frequency is chosen to be $f_s/24$; this greatly reduces the downconversion of shaped out-of-band quantization noise into the signal band. For this measurement, the FIR-MASH output is decimated to 48 KS/s on an FPGA, with the objective of achieving better FFT resolution. We see that chopping is effective in modulating flicker noise out of the signal band.

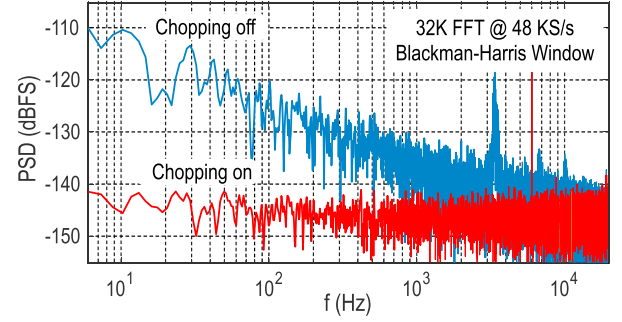
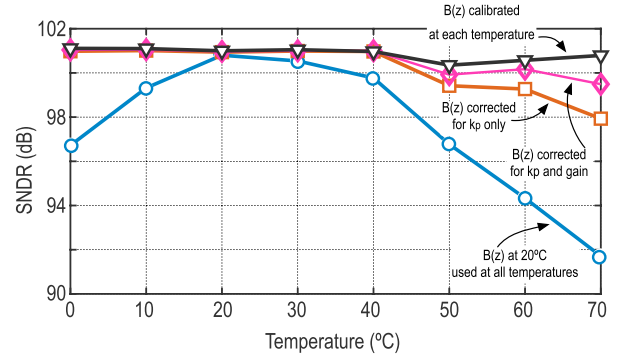


Fig. 11. In-band PSD with chopping on/off.

Fig. 12. Measured SNDR for -0.95 dBFS, 6-kHz input as a function of temperature, with different calibration strategies.

A. Calibration Performance Over Temperature

Recall that the CT $\Delta\Sigma$ M output is reconstructed according to $V(z) = F(z)A(z)V_1(z) + B(z)V_2(z)$. As discussed in Section III, it is sufficient to adjust the two-tap FIR filter $B(z)$ to cancel the quantization noise of MOD1. In this section, we show experimental results where $B(z)$ is derived using foreground calibration at room temperature, and the coefficients so obtained are subsequently adjusted for variations in k_p and finite integrator gain. For these measurements, the test board was placed inside a temperature chamber and data captured every 10 °C from 0 °C to 70 °C.

Fig. 12 shows the measured SNDR as the temperature is varied, with an input amplitude of -0.95 dBFS. When $B(z)$ derived at room temperature is used at all other temperatures, we see that the SNDR degrades to about 92 dB at 70 °C. When $B(z)$ is modified to account for k_p variation (obtained from foundry models for resistor temperature coefficient), the SNDR at 70 °C improves to 98 dB. When both k_p and finite gain are used to appropriately modify the $B(z)$ obtained at room temperature, the SNDR improves to 99.5 dB. From these results, it appears as if the simple calibration technique can achieve acceptable performance across temperature. As mentioned earlier, the primary aim of this project was to test the efficacy of FIR-MASH architecture. An improved design would incorporate an RC oscillator, whose frequency of oscillation can be used to infer k_p . Calibration of finite integrator gain can be avoided altogether by using an extra gain stage in the OTA, making A_0 so large as to be inconsequential.

TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART $\Delta\Sigma$ CONVERTERS

Reference	[14]	[15]	[16]	[17]	[18]	[19]	[20]	[21]	[8]	This work
BW (kHz)	24	24	24	20	25	20	24	20	24	24
Technology Node (nm)	180	65	28	160	65	160	65	160	180	180
Supply (V)	1.8	1.1	3.3/1.0	1.8	1.0	1.6	1.2	1.8	1.8	1.8
Power (μ W)	280	121	1130	1650	800	390	68	618	280	265
Peak SNDR (dB)	98.2	85	98.5	98.3	95.2	91.3	94.1	106.4	98.5	100.9
Peak SNR (dB)	99	87	100.6	104.4	100.1	-	-	108.1	99.3	101.7
DR (dB)	103	88	100.6	107.5	103	103.1	98.2	108.5	103.6	104
SFDR (dB)	106	90	102.6	98.3	100	98.4	107	113	107.6	108
Area (mm ²)	1	0.6	0.02	0.16	0.25	0.21	0.14	0.27	1	0.64
Chopping Freq. (kHz)	-	-	-	-	47	-	-	-	256	256
FoM _{Schreier}	177.5	168	172	169.1	170	168.4	179.5	181.5	177.8	180.5/179.6 [†]

$$\text{FoM}_{\text{Schreier}} = \text{SNDR}_{\text{max}} + 10 \log \left(\frac{\text{BW}}{\text{Power}} \right) \quad (\dagger \text{Schreier FoM including } 60 \mu\text{W for the noise-canceling filters})$$

The power dissipation of the noise-cancellation filters, estimated from digital synthesis and place-and-route, is about 60 μ W.

Table I gives a performance summary and compares our design with state-of-the-art audio CT $\Delta\Sigma$ Ms. We see that our work achieves a performance that is competitive with respect to the state of the art. Compared with [4], the Schreier FoM is higher by about 3 dB, mostly on account of the increased MSA of the 1-X FIR-MASH architecture.

VI. CONCLUSION

This article introduced the 1-X FIR-MASH CT $\Delta\Sigma$ M architecture, whose objective is to use a first-order front end to achieve a $\Delta\Sigma$ converter whose MSA is close to full scale. FIR feedback was shown to be very beneficial in the context of a MASH converter, where it can be used to filter the error of the first stage before being digitized by the second. This reduces the peak-to-peak swing of the error processed by the second stage, as well as its high-frequency content. This benefit is apart from FIR feedback's other well-recognized advantages, such as jitter-sensitivity reduction, linearity improvement, and attenuation of chopping artifacts. Measurement results from a 1-X FIR-MASH test chip designed in a 180-nm CMOS technology show that the CT $\Delta\Sigma$ M achieves 100.9-dB peak SNDR in a 24-kHz bandwidth. Though only single-bit quantizers are used, the MSA is about -0.65 dBFS. While demonstrated in a low-bandwidth converter, the FIR-MASH technique appears to be a promising approach even at higher bandwidths.

APPENDIX

In this appendix, we describe a practical technique used to determine the taps of the noise-canceling FIR filter $B(z)$. As discussed earlier, if only in-band cancellation is desired, $B(z)$ can be restricted to two taps. The tap values are those that minimize the noise power of v in the signal band. They are determined as shown in Fig. 13.

The FFTs of $v_2[n]$ and its one-sample delayed version $v_2[n-1]$ are computed after windowing the sequences with an appropriate window function, denoted by $w[n]$. Windowing is necessary to avoid leakage of out-of-band quantization noise into the signal band. The in-band bins are collected, and those bins corresponding to the input signal and dc are discarded.

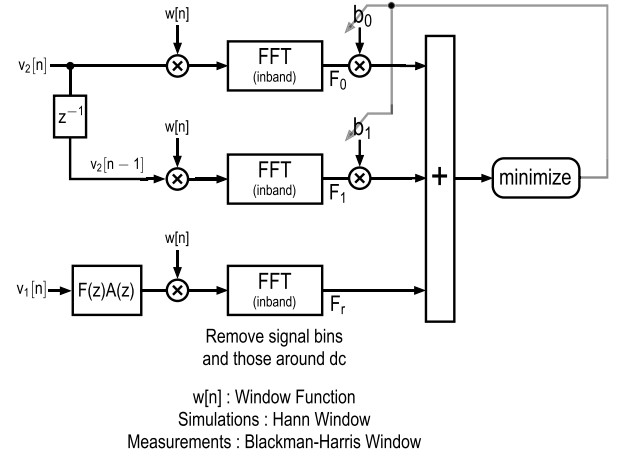


Fig. 13. Determining the coefficients of the two-tap FIR filter $B(z)$ by minimizing in-band quantization noise.

The resulting vectors are denoted by F_0 and F_1 . v_1 , after being filtered with $F(z)A(z)$, is also processed in the same manner, and the resulting in-band FFT is denoted by F_r . Minimizing in-band noise of the output of the MASH CT $\Delta\Sigma$ M is equivalent to finding the least-squares solution to

$$b_0 F_0 + b_1 F_1 = -F_r \quad (12)$$

where b_0 and b_1 must be real. In the matrix form, this system of equations can be written (assuming F_i 's are column vectors) as $\mathbf{F}\mathbf{b} = -F_r$, where

$$\mathbf{F} = [F_0 \ F_1] \text{ and } \mathbf{b} = [b_0 \ b_1]^T. \quad (13)$$

The vector $\mathbf{b}$ that minimizes the norm of $\mathbf{F}\mathbf{b} + F_r$ (or equivalently, the power of $b_0 F_0 + b_1 F_1 + F_r$ in the signal band) can be found by solving the normal equations

$$\begin{bmatrix} \text{Re}(\mathbf{F}) \\ \text{Im}(\mathbf{F}) \end{bmatrix} \mathbf{b} = - \begin{bmatrix} \text{Re}(F_r) \\ \text{Im}(F_r) \end{bmatrix}. \quad (14)$$

Straightforward algebra shows that the solution is given by

$$\mathbf{b} = -[\text{Re}(\mathbf{F}^H \mathbf{F})]^{-1} \text{Re}(\mathbf{F}^H F_r). \quad (15)$$

where $\mathbf{F}^H$ denotes the Hermitian transpose of $\mathbf{F}$.

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